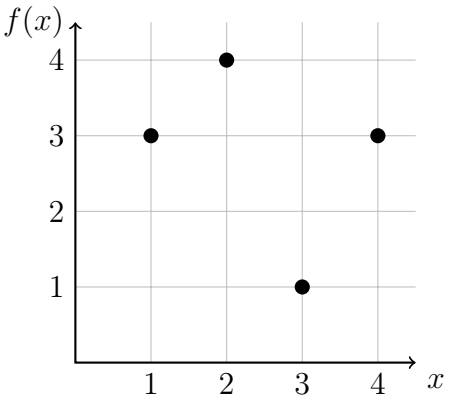
# Discrete Mathematics — Tutorial Sheet 06 — Functions

BSc (H) in Comp Sci, BSc (H) in Comp Foren

**Question 1**

Consider the function $f : \{1, 2, 3, 4\} \to \{1, 2, 3, 4\}$ given by the graph below.




**(a)** Is $f$ injective? Explain.

**(b)** Is $f$ surjective? Explain.

**(c)** Express the function using a look-up table and a digraph.

**Question 2**

Consider the function $f : \{1, 2, 3, 4, 5\} \to \{1, 2, 3, 4\}$ given by the table below:

| $x$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $f(x)$ | 3 | 2 | 4 | 1 | 2 |

**(a)** Is $f$ injective? Explain.

**(b)** Is $f$ surjective? Explain.

**Question 3**

Let $f : A \to B$ be some function. Suppose $5 \in B$. What can you say about $f^{-1}(5)$ if you know,

**(a)** $f$ is injective? Explain.

**(b)** $f$ is surjective? Explain.

**(c)** $f$ is bijective? Explain.